A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 05

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Do not write outside the box

Question	Marking guidance	Marks
01.1	**Marks shared** Up to 6 marks in total - share them between the 4 points below according to how completely each one is made. **up to 2 Organic advantages:** often waste products, locally available and cheap – add humus improving structure and water retention – increase soil biota – slow release reduces leaching **up to 2 Organic disadvantages:** nutrient composition cannot be controlled – released slowly so needs long-term planning – bulky with high water content so expensive to transport – cannot be applied to a growing crop – odour / pathogens **up to 2 Inorganic advantages:** composition controlled to match crop needs – nutrients released rapidly – concentrated and easy to apply **up to 2 Inorganic disadvantages:** high energy (fossil fuel) manufacture (Haber process) – no organic matter so humus and biota decline – some toxic to worms – high solubility so leached causing eutrophication – raw materials (phosphate rock) finite **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **humus** → dark, decomposed organic matter in soil • **leaching** → nutrients or ions washed down out of the soil / washed away by water • **eutrophication** → nutrient enrichment of water leading to algal growth **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	6
01.2	**Marks shared** Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2** Crops grown in a circulating nutrient solution (inert medium) in greenhouses – nutrient concentrations monitored and adjusted so nutrients are never limiting – abiotic factors controlled **up to 2** Roots are small (all in contact with nutrients) so more growth goes into the harvested crop – no soil-borne pathogens or weeds – crop stays fresh with roots **up to 2 Disadvantages:** high inputs of nutrients and energy (heating, lighting, pumping) – high level of technical knowledge – only economic for high-value crops **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 310-312; AQA spec 3.5.1.1.	

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Question	Marking guidance	Marks
02.1	**1 mark each** 1 mark for each of the 4 points below. **1** Silver carp eat phytoplankton, bighead carp zooplankton, grass carp and tilapia vegetation, common carp bottom-feed on molluscs / insects, catfish or perch eat small fish **1** Species occupy different niches / trophic levels so they do not compete and all food sources are used **1** Bottom feeders re-suspend nutrients boosting plankton / plant growth **1** Predators control over-breeding species (tilapia) so remaining fish grow larger – total yield exceeds any single species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	4
02.2	**1 mark each** 1 mark for each of the 4 points below. **1 IMTA:** fed species (salmon, shrimp) alongside inorganic extractive species (seaweeds absorbing ammonia and phosphate) and organic extractive species (filter-feeding mussels eating waste particles / plankton) **1** Wastes of one species become nutrients for another so less pollution and extra harvests **1 Aquaponics:** aquaculture water rich in nutrients / organic matter feeds hydroponic salad crops (lettuce, herbs, watercress) **1** Plants remove the nutrients that would cause deoxygenation and cleaned water returns to the fish (tilapia, carp, catfish) – rice-fish paddies similar **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming • **deoxygenation** → loss of dissolved oxygen from the water **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	4
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Bluefin tuna do not breed well in captivity so young fish are caught from the wild and fattened in sea cages **1** Production still depends on (and removes juveniles from) the wild population – carnivores need large fish inputs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 357-358; AQA spec 3.5.2.3.	
03.1	**1 mark each** 1 mark for each of the 4 points below. **1** Trees (hazel, oak) are cut to ground level every 3-30 years and allowed to regrow from the stump **1** About 0.5 ha cut each year on a rotation (eg 8 years) so the wood is a patchwork of areas of different ages **1** Each age has its own community: newly cut areas get high light for wildflowers and butterflies – older areas give dense cover for nesting birds / dormice **1** Pollarding cuts above browsing height so deer / livestock cannot eat the regrowth **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	4
03.2	**1 mark each** 1 mark for each of the 4 points below. **1** Wolves (top predator, exterminated by 1926) reduced deer populations and changed their behaviour **1** Fewer young trees eaten so trees and riverside vegetation recovered **1** More trees allowed beaver populations to increase – beaver dams increased wetland area **1** Wetland species (birds, fish, invertebrates) became more common – overall biodiversity rose, though wolves cause problems for livestock farmers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	4
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Undesirable species (introduced predators / competitors, or natural succession species such as scrub, conifers in the New Forest heathland) reduce survival of the target species **1** Removal restores suitable conditions (eg heathland for sand lizards and smooth snakes) or reduces predation / competition **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **succession** → the gradual change in the community over time **Do not credit** • 'the climax community never changes' - it is stable only while the conditions stay the same	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 47-49; AQA spec 3.1.3.3-3.1.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**1 mark each** 1 mark for each of the 2 points below. **1** About 10% of land is already cultivated and 25% grazed **1** Most remaining land is too dry, wet, cold, hot or steep, is covered in ice, or has no soil **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Land becomes too expensive for local people to rent / is used to grow export crops for more affluent societies **1** So people with less money lose access to the land and food they used to produce **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** It uses natural processes to supply nutrients (legumes, recycled organic matter, soil biota), control pests and provide ecological services **1** So that pesticides and artificial fertilisers become unnecessary rather than just being banned **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	2

Do not write outside the box

Question	Marking guidance	Marks
04.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Unsustainable features:** reliance on chemical pesticides (resistance, non-target impacts, bans), non-renewable rock phosphate, fossil-fuel-intensive nitrate manufacture, over-exploitation of rivers / groundwater, salinisation, loss of soil organic matter, greenhouse gases from fuel, rice and livestock, loss of species providing ecosystem services • **Pest control:** cultural methods (weeding, mulching, crop rotation, barrier crops, culling), biological control, predator habitats, polyculture / companion crops, integrated control, reduced antibiotic use • **Nutrients:** recycling organic matter, crop rotation, cultivation of legumes, conservation of soil biota, reduced artificial fertiliser especially nitrate, low tillage • **Water:** low water-use crops, reservoirs and aquifer recharge, drip irrigation rather than sprays • **Soil and carbon:** low-tillage farming to maintain organic matter, reduced machinery, renewable energy, drought-tolerant rice varieties so fields are drained earlier, high-carbohydrate ground feed for cattle reduces methane • **Biodiversity:** conserve habitats of crop wild relatives, seed banks, retain hedgerows, ditches, ponds, woodland – agri-environment schemes • **Evaluation:** growing population (perhaps 12 billion) needs more food – lower trophic level diets – obstacles - cost, need for support, gaining acceptance of new techniques – learning from past land degradation (fertile crescent, US mid-west) • **Judgement:** no single strategy sufficient – combination of ecological principles and new technology – justified conclusion **Also accept**	9

Do not write outside the box

Question	Marking guidance	Marks
	• Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **salinisation** → salt building up in the soil / soil becoming salty • **biological control** → using a natural predator, parasite or pathogen of the pest **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	
	Model answer - Level 3 (8 marks)	
	Present agriculture relies on inputs that cannot last: chemical pesticides to which pests evolve resistance and which harm non-target species, rock phosphate that is non-renewable, nitrate fertiliser made with fossil-fuel energy, and irrigation that over-draws rivers and groundwater and salinises soil, while ploughing loses soil organic matter and livestock, rice and machinery release greenhouse gases. Each can be replaced by strategies based on ecological principles. Pests are controlled culturally by weeding, mulching, rotation, barrier crops and culling, by biological control and predator habitats, by polyculture and companion crops, and by integrated control that uses chemicals only as a last resort. Nutrients come from recycled organic matter, rotation with legumes and conserved soil biota, so artificial fertiliser, especially nitrate, can be reduced. Water is saved with low water-use crops, reservoirs and aquifer recharge and drip irrigation; soil carbon is kept by low tillage and less machinery, with renewable energy for what remains; methane falls with rice varieties that allow earlier drainage and with ground, carbohydrate-rich cattle feed. Habitats of crop wild relatives, seed banks, hedgerows, ditches, ponds and woodland and agri-environment schemes protect the biodiversity that supplies these services.	
	Food demand will rise with a population that may reach 12 billion, there is little new land, and imports can reduce food availability where it is grown, so yield cannot be ignored. Some strategies cost more or need farmers to accept new techniques, and past land degradation from the fertile crescent to the American mid-west shows what happens when they are not adopted. No single strategy suffices; a combination of ecological principles with new technology is the only route to feeding more people without exhausting the resources food depends on.	
	Weaker answer - Level 1 (2 marks)	
	To make agriculture sustainable farmers should use fewer chemicals. Organic farming does not use pesticides or artificial fertilisers. Crop rotation and manure keep the soil fertile. Farmers should not waste water. Hedges should be kept for wildlife. This would make farming more sustainable but it might produce less food.	
	What makes the difference	
	• Level 1 gives general advice; Level 3 identifies each unsustainable feature and matches it with specific strategies, showing understanding of why they work.	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 3's evaluation considers population growth, land limits, cost, adoption and historical evidence; Level 1's is a single clause.	
	• Note that Level 3 avoids defining organic farming as 'not using chemicals' - the textbook stresses it is the positive use of natural processes.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 335-337; AQA spec 3.5.1.5.	
05.1	**1 mark each** 1 mark for each of the 3 points below. **1** The range of an environmental factor within which a species / population can survive (survival highest at the optimum, falling towards the limits) **1** X survives from about pH 5.5 to 8.5 (wide range) – Y only from about pH 6.0 to 8.0 (narrow range) **1** Both have the same optimum (pH 7.0) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **range of tolerance** → the range of conditions in which a species can survive	3
05.2	**1 mark each** 1 mark for each of the 3 points below. **1** X: 60% survive at pH 6.0 whereas only 5% of Y survive **1** X has a larger gene pool / wider range of tolerance so some individuals are adapted to the new conditions **1** Survivors breed and their offspring are adapted so the population rebuilds – Y is likely to die out locally **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population • **range of tolerance** → the range of conditions in which a species can survive	3
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Reveals the habitat requirements / which abiotic factor is near the edge of tolerance **1** Management can make small beneficial changes to that factor (eg liming, shading, water levels) to keep conditions within the range and increase survival **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **range of tolerance** → the range of conditions in which a species can survive • **abiotic** → non-living / physical factors	2

Do not write outside the box

Question	Marking guidance	Marks
05.4	**Any 2 of 5** 1 mark each for any 2 of the 5 points below, in any order, maximum 2. **1** Food supply **1** Pollinators **1** Seed dispersal species **1** Disease / pathogens **1** Predators / competitors **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biotic** → living factors / effects of other organisms **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 72-73; AQA spec 3.1.3.1.	
06.1	**1 mark each** 1 mark for each of the 2 points below. **1 Moral belief:** protecting the environment is the 'right' thing to do regardless of benefit to people **1 Enlightened self-interest:** protecting the environment is done because it causes fewer difficulties (for people) than not doing so **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Total impact = population size x the individual (per capita) impact of each person **1** Per capita impact rises with affluence / access to energy, so both a larger population and a more affluent population increase the total impact **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2

Do not write outside the box

Question	Marking guidance	Marks
06.3	**Any 1 of 3** 1 mark each for any 1 of the 3 points below, in any order, maximum 1. **1** Slash and burn agriculture **1** Fishing **1** Discharge of sewage / biodegradable pollutants **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **biodegradable** → broken down by decomposers **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370; AQA spec 3.6.1.	
07.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** Environmental DNA (eDNA) analysis	1
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Transmitters are too heavy for small birds / their batteries would not last for a long-term study **1** Geolocator tags are small and only record time and light levels – latitude and longitude for each day are worked out when the bird is recaptured and the data downloaded **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Each separate (regional) population has its own unique gene pool **1** DNA from seized ivory can be compared with the database to identify where the elephant came from, so poaching hotspots / trade routes can be targeted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 404-405; AQA spec 3.1.2.3.6.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**Marks shared** Up to 5 marks in total - share them between the 4 points below according to how completely each one is made. **up to 2** A: coppice rotation creates a patchwork of different ages / light levels - clearings with wildflowers for butterflies, dense regrowth for nesting birds – oak supports many invertebrates **up to 2** B: same tree species but the closed canopy shades the floor so few flowering plants, fewer invertebrates and butterflies – fewer niches **up to 2** C: non-indigenous single-species, single-age conifers support few native invertebrates – dense shade – no clearings or dead wood – needles acidify soil **up to 2** Habitat diversity / structural variety increases the range of niches and hence species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **biodiversity** → the variety of life: habitat, species and genetic diversity	5
08.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1 Designated protected areas:** SSSIs, NNRs (Sherwood Forest), SACs (Epping Forest), National Parks **1 Legal protection of ancient woodland:** planning permission refused for developments destroying irreplaceable habitats (2012 NPPF) **1 Conservation management:** coppicing, creating clearings, planting mixed native species, leaving dead wood, controlling deer **1** Planting new native woodlands linking fragments (corridors) – Woodland Trust / National Trust ownership **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** No species is completely dominant and food webs have many alternative links **1** So a change in the population of one species has a relatively small impact on the whole community **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-52; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**1 mark each** 1 mark for each of the 2 points below. **1** Island communities evolved from few colonising species in isolation, often without mammal predators or large herbivores **1** So indigenous species lack adaptations (eg ground-nesting birds, flightlessness) to survive introduced cats, rats, pigs, goats **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
09.2	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Competitor:** grey squirrel (North America) out-competes red squirrel - digests acorns, larger, wins nest sites – rhododendron shades ground and releases toxins preventing native regrowth **2 Predator:** American mink (escaped fur farms) reduced water vole range – cane toads in Australia – cats / rats / pigs on islands eating ground-nesting birds – Nile perch in Lake Victoria ate cichlids to extinction **2 Pathogen:** squirrel pox virus carried by grey squirrels kills red squirrels – crayfish plague carried by signal crayfish kills white-clawed crayfish – Dutch elm disease, ash dieback, sudden oak death imported on plants / soil **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	6
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** Closely related introduced species (sika deer; domestic cats) cross-breed producing fertile hybrids **1** The natural gene pool is changed / diluted with genes it would not naturally contain, so the pure indigenous species may disappear **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 24-25; AQA spec 3.1.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Water:** catchment management, conservation, metering, greywater, re-use of treated effluent, aquifer recharge, sustainable urban drainage - largely cyclical already – limits in arid regions • **Minerals:** exploiting low-grade ores with low-temperature processes (bioleaching), extending product lifespans, repair / re-use / repurpose / recycle, design for disassembly, reduced packaging – limits: dispersal, alloys, energy of collection • **Waste management:** repair, re-use, refurbish, recycle, composting and anaerobic digestion returning nutrients, minimal landfill, energy from residual waste – EU targets on packaging, batteries, WEEE • **Pollution control:** shift from post-production treatment to non-release - fuel desulfurisation not FGD, fuel cells / electric vehicles not combustion, industrial symbiosis (Kalundborg: waste heat, gypsum, fly ash, yeast used by neighbours) – limits for legacy pollution • **Energy:** renewables, storage, energy density technologies, low-temperature manufacturing, CHP – limits of intermittency, transport fuels • **Food:** natural processes for nutrients (legumes, organic matter), pest control (IPM), soil biota, agri-environment schemes, IMTA aquaculture – tension with maximum yield for a growing population • **Land uses supporting ecosystems:** urban green infrastructure, gardens, permeable surfaces, habitat in redeveloped docks / lagoons • **Judgement:** high applicability in some sectors (water, waste), partial in others (minerals, energy for transport) – needs policy support **Also accept**	9

Do not write outside the box

Question	Marking guidance	Marks
	• Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **aquifer** → a permeable rock that stores groundwater / underground water store • **sustainable** → meeting present needs without stopping future generations meeting theirs • **desulfurisation** → removing sulfur dioxide from flue gases with lime or limestone **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	
	Model answer - Level 3 (9 marks)	
	Taking water, waste and pollution control: water supply is already close to circular, because the hydrological cycle returns what is used. Catchment management, conservation, metering, grey-water re-use, re-use of treated effluent, aquifer recharge and sustainable urban drainage close the loop further, with the main limit in arid regions where abstraction exceeds recharge. Waste management can apply the principles almost completely: repair, re-use and refurbishment extend product life, recycling returns materials, composting and anaerobic digestion return nutrients and yield biogas, residual waste supplies energy and landfill is minimised, driven by EU targets on packaging, batteries and WEEE; the limits are mixed materials that cannot be separated and the energy cost of collection.	
	Pollution control changes character under circular principles: instead of treating wastes after production - flue-gas desulfurisation, dispersants - the aim becomes non-release, through fuel desulfurisation, fuel cells and electric vehicles, and industrial symbiosis, as at Kalundborg where one plant's waste heat, gypsum, fly ash and yeast become neighbours' raw materials. Land use can support ecosystems with green infrastructure, gardens, permeable surfaces and habitat in redeveloped docks.	
	Applicability is high for water and waste because natural cycles already exist to copy; it is partial for pollution control because many processes still depend on combustion, and it needs policy support - targets, producer responsibility, planning - to overcome the economics of linear systems. The principles can be applied widely, but not evenly.	
	Weaker answer - Level 1 (3 marks)	
	The circular economy means reusing and recycling things instead of throwing them away. For water this means saving water and reusing grey water. For waste management it means recycling more and putting less in landfill. For pollution control it means not producing pollution in the first place. The circular economy can be applied to all of these to make them more sustainable.	
	What makes the difference	
	• Level 1 gives one idea per area; Level 3 gives several specific measures for each and names the limit in each case.	
	• Level 3 uses a case study (Kalundborg) and policy examples (EU targets) to show application, not just principle.	

Do not write outside the box

Question	Marking guidance	Marks
	• 'Evaluate the extent' - Level 3 compares applicability across the three areas and explains the differences; Level 1 says it applies to all.	
10.2	1 mark for the point below. **1** Locating industries so that the wastes of one (heat, gases, by-products) become the raw materials of another, mimicking natural cycles (eg Kalundborg eco-industrial park) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **symbiosis** → a close relationship in which both species benefit	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 375-379; AQA spec 3.6.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Resources:** wood, fibres, oils, fuels, food – new food species (morama bean, yeheb, perennial maize, Kernza, bison, eland) and their adaptations • **Biomimetics:** wing-tip feathers, whale tubercles, shark skin, termite mounds, bird bones, gecko adhesion, burrs, lotus, spider silk • **Medicines:** poppies, cinchona, willow, yew (taxol), Mexican yam, sponges (AZT), sharks – few species yet studied – physiological research (marsupials, hippo, sponges) • **Genetic resources:** crop wild relatives, Vavilov centres, gene pools and inbreeding • **Ecosystem services:** atmospheric regulation, hydrological cycle, biogeochemical cycles, soil maintenance – interspecies relationships - food, pollination (Darwin's orchid), seed dispersal, habitat provision • **Evaluation:** economic vs intrinsic / moral arguments – services unnoticed because of dynamic equilibrium – unknown future value – justified conclusion	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	

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Question	Marking guidance	Marks
11.2	**Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Energy flow:** autotrophs and heterotrophs – losses by reflection, transmission, respiration – 100 000 to 1 000 units – few food chains beyond four levels – plant food most productive – grazing animals where crops impossible – pigs on wastes • **Energetics:** intensive vs extensive, law of diminishing returns, energy subsidies (fertiliser, pesticides, irrigation, machinery, drying, processing, transport), energy ratios, food conversion ratios, dependence on fossil fuels • **Control of genetics:** asexual reproduction / cuttings / cloning, selective breeding and breeds, cross-breeding and hybrid vigour, AI and embryo transfer • **GM:** Roundup Ready soya, Bt crops, golden rice, omega-3 oilseed rape – advantages – risks (gene transfer, antibiotic markers, allergies, costs to LEDC farmers, loss of local diversity)	25
	• **Sustainability links:** energy dependence, trophic level of diet, feeding 12 billion people • Justified conclusion on the balance between productivity, efficiency and sustainability *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-328; AQA spec 3.5.1.2.	

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